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SECTIONS:

	STUDENT OUTCOMES	COURSE LEARNING OUTCOMES	WEIGHT OF SECTION /100	MARKS OBTAINED	REMARKS
Abstract	1,6	5	5		
Introduction	1,6	5	10		
Analytical models	1,6	5	20		
Finite element analysis	1,6	5	25		
Discussion	1,6	5	15		
Conclusion	1,6	5	10		
Report Format And References			15		
TOTAL:(Out of 100)					

Instructions:	Student Outcomes:	
- No Late Submissions accepted. - Please include the equations, diagrams, plots, and figures in the report. - All the sources of data, equations, diagrams, plots, and figures used must be cited. - Please provide the references and cite each reference in the text. - All references must be cited in APA format. - Please submit the pdf soft copy of the report via email . - The file uploaded should be named as [MENG376-Project-Group_No] Files with any other name or format will be disregarded.	1	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
	2	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
	3	an ability to communicate effectively with a range of audiences
	4	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
	5	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
	6	an ability to develop and conduct appropriate experimentation, analyze, and interpret data, and use engineering judgment to draw conclusions
	7	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Abstract

This study examines a static analysis of a mechanical shaft using Finite Element Analysis (FEA) focusing on stress distribution and deformation under static loading conditions. The shafts are fundamental components in mechanical systems, responsible for transmitting torque and rotational motion in applications such as automotive powertrains, industrial machinery, and energy systems. An exact 3D model of the shaft is made using Ansys Speos then ANSYS Workbench is used to perform static structural simulations, giving the results in terms of von Mises stress, total deformation, and factor of safety. This study shows how variations in material selection, shaft geometry, and applied loads influence performance and potential failure points. High-stress concentration regions are shown, offering important points for design optimization and structural integrity enhancement. The analysis follows the ASME design standards and builds upon basic knowledge from authors such as Shigley, Budynas, and Mott. The results focus the importance of using simulation tools in the early design phase to improve reliability, reduce material costs, and extend component lifespan. This research helps to the broader understanding of shaft mechanics and supports the development of safer, more efficient mechanical systems.

Keywords: Static analysis, Shaft design, Finite Element Analysis (FEA), Stress distribution, ANSYS simulation, Mechanical failure prevention

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Introduction

The static analysis of shafts is crucial for understanding their performance under different loading conditions, especially in electromechanical systems. Recent studies have delved into innovative configurations such as electromagnetic working shafts (EMWSh) and electro-electromagnetic working shafts (EEMWSh), incorporating capacitors into rotor circuits to enhance static performance [1]. Shafts used in mining environments, particularly near salt rock formations, also demand lightweight yet robust structural supports to maintain shaft integrity under static conditions [2]. Condition monitoring techniques, especially those based on bearing diagnostics, have been developed to improve the reliability assessment of cardan shafts by detecting early signs of failure under static loads [3]. The Finite Element Method (FEM) remains one of the most prominent tools in static shaft analysis due to its ability to simulate various load scenarios and provide detailed insight into stress distributions [4]. CAD and CAM systems have increasingly been adopted in engineering education to foster better understanding of static and dynamic shaft behaviors, highlighting the practical value of simulation technologies [5]. In agricultural applications, universal joint shafts have been studied using temperature-based reliability assessments to evaluate static conditions in bearing assemblies [6]. Expanding on FEM applications, researchers have shown how different loading variants influence the static response of shafts, confirming the method's versatility and accuracy [4]. Offshore structures, such as piles, benefit from full-scale static axial testing to evaluate shaft response in geotechnically complex environments like chalk and tills [7].

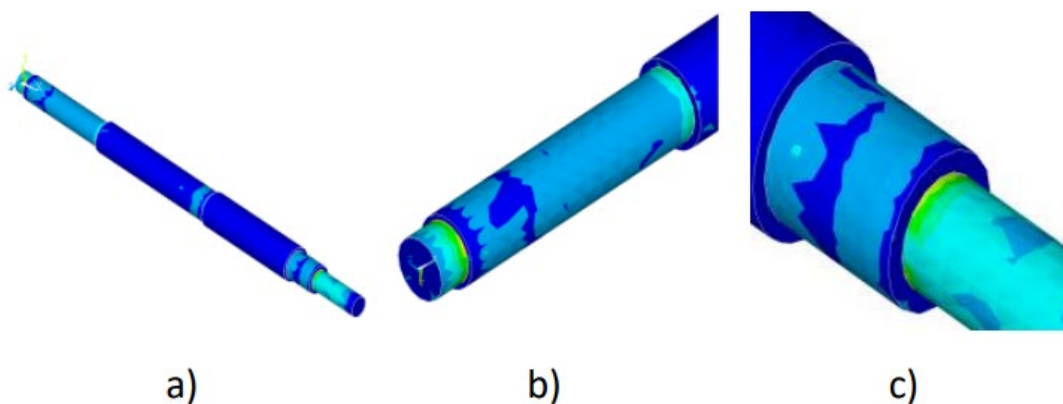


Figure 1 stress contours [4]

In the automotive sector, propeller shaft optimization using structural analysis plays a vital role in ensuring performance efficiency under static conditions[8]. Capacity-based shaft design has also been evaluated, emphasizing how static loading influences shaft dimensioning and material selection [9].

Shafts in mining operations often face challenging stress-strain environments, necessitating accurate computer simulations for safer structural designs[10]. Composite shaft performance under static and dynamic loads is significantly influenced by the fiber type used, highlighting material choice as a key design consideration[11]. Hybrid shafts combining aluminum and composites have also undergone dynamic analysis, providing insights into the static characteristics that precede failure[12]. Closed-form solutions for shaft-gear systems present an analytical pathway to understand static equilibrium without relying solely on numerical tools[13]. Defect analysis in rotating shafts shows that static analysis is foundational in understanding how material discontinuities evolve into critical failures[14]. The prediction of stress states in vertical mine shafts through FEM remains essential for maintaining shaft lining integrity, especially in saliferous environments[15]. Surface defects such as scratches on shaft surfaces also impact static behavior, particularly in bearing-supported configurations[16].

Advanced turbofan designs have prompted the study of static shaft combustion characteristics, illustrating cross-disciplinary applications of shaft mechanics[17]. Theoretical studies have complemented practical analysis, offering mathematical insight into shaft bending and deflection under static loads[18]. Foundational knowledge of FEM continues to shape how shafts are analyzed, allowing engineers to model complex shaft geometries under static conditions with high accuracy[19]. Agricultural shaft assemblies have also benefited from improved reliability models grounded in static evaluation of bearing performance[20].

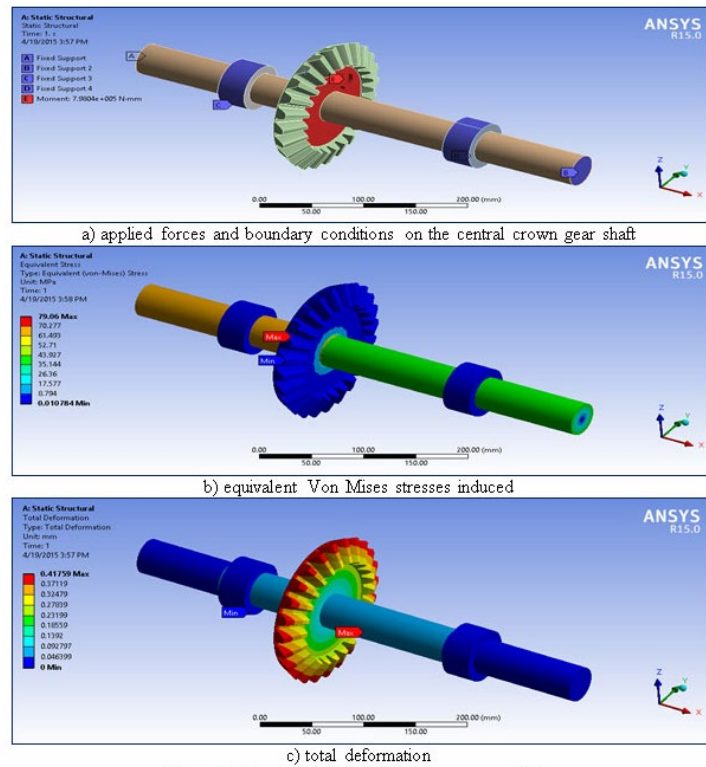


Fig. 9: FEA results of the central crown gear shaft

Figure 2 Shaft boundary conditions [21].

Innovations in computational methods such as the Finite Cell Method allow high-resolution shaft analysis even for CAD-derived geometries[22] . In energy storage, mine shafts are repurposed into large-scale mechanical batteries, demanding careful static evaluations of structural integrity [23]. Static evaluation of pump shafts provides critical insights for centrifugal machinery performance, especially under prolonged stationary conditions[24]. FEM-based performance algorithms have also been successfully applied in brake mechanisms, providing transferable techniques for static shaft analysis[25].

Vibration control studies show that static shaft deflections can influence the onset of nonlinear oscillations, underlining the need to understand static conditions as a prerequisite to dynamic control [26]. Finally, static torque stabilization has been explored in crank mechanisms, linking mechanical balance with structural integrity of the shaft system[27] .

Theory

The application of Finite Element Analysis (FEA) in the static analysis of mechanical shafts provides critical insights into stress distribution, deformation, and design optimization. Static loads such as torque, bending moments, and transverse forces often arise from mounted components like gears and pulleys, making accurate modeling essential for safe design [28]. Shigley and Budynas [29] emphasize the necessity of considering fillets and keyways due to their tendency to create stress concentrations. ANSYS Workbench, as outlined by Moaveni [30] and Lawrence [31], enables precise simulation of these effects through its powerful meshing and boundary condition tools. Research by Ašonja et al. [4] demonstrates the effectiveness of FEA in predicting shaft behavior under varying load conditions, showing good agreement with analytical results. Deshmukh and Kulkarni [32] similarly validated the stress distribution in pump shafts using ANSYS, highlighting the importance of proper constraint and load applications. Kadam and Kulkarni [33] extended this to spline shafts, verifying results through experimental data. Khan and Singh [34] conducted a comparative study on composite shafts, showing how material selection significantly influences deformation and factor of safety. Standards such as DIN 743 [35] and ASME B106.1M [36] offer guidelines for modeling real world shaft loading, including bearing forces and safety considerations. These studies collectively demonstrate the robustness of FEA in static shaft analysis and the value of ANSYS as a predictive and diagnostic tool.

Model

Model Example – A Cantilevered Propeller Shaft (Small/Medium Scale)

Application Scenario:

Electric fan, drone propeller or small water pump

The shaft has a roller attached at the free end.

The shaft is fixed to motor housing at the one end (Fixed Support)

Table 1 Average Shaft Dimensions (Steel or Aluminum)

Feature	Typical Range	Description
Length (L)	100 – 300 mm	Longer shafts deflect more easily.
Diameter (d)	6 – 20 mm	Depends on torque and load.
Cross-Section	Circular (Ø)	Ideal for bearings and balancing.
Material	AISI 1045 Steel / 7075 Aluminum	Provides good stiffness and strength.
Torque (approx.)	0.5 – 5 Nm	Common for small propeller loads.

Key Design Considerations

$$\delta_{\max} = \frac{FL^3}{3EI}$$

(1)

F: Force at the Propeller

L: Shaft length

E: Young’s Modulus (210 GPa for steel)

I: Area moment of inertia

$$\tau = \frac{Tr}{J}$$

(2)

T: Applied Torque

r: Shaft radius

J: Polar moment of Inertia

Chosen Model Dimensions

Diameter = 10mm

Length = 150mm

Material = Structural Steel (AISI 1045)

$E = 210 \times 10^9 \text{ Pa}$

$I = 4.91 \times 10^{-11} \text{ m}^4$

Support Type = Cantilever Support

Use Case = Small fan or drone propeller

We will model the propeller as a force at the free end.

Propeller weight and gravity is neglected.

Torque from propeller is neglected we'll only do analysis about the propeller attached at the end of the shaft.

Table of Properties Row 2: Density		
	A	B
1	Temperature (C)	Density (kg m ⁻³)
2		7850

Properties of Outline Row 13: Structural Steel			
	A	B	C
1	Property	Value	Unit
2	 Density	7850	kg m ⁻³
3	 Isotropic Secant Coefficient of Thermal Expansion		
5	 Isotropic Elasticity		
6	Derive from	Young's Modulus and...	
7	Young's Modulus	2E+11	Pa
8	Poisson's Ratio	0,3	
9	Bulk Modulus	1,6667E+11	Pa
10	Shear Modulus	7,6923E+10	Pa
11	 Strain-Life Parameters		
19	 S-N Curve	 Tabular	
23	 Tensile Yield Strength	2,5E+08	Pa
24	 Compressive Yield Strength	2,5E+08	Pa
25	 Tensile Ultimate Strength	4,6E+08	Pa
26	 Compressive Ultimate Strength	0	Pa

Figure 1 Structural Steel is chosen from Engineering Data Source

Shown in Figure 1 the First step we open our ANSYS workbench and then we select static structural from left and then we open our system. First, we need to select material to be assigned later. We click engineering data and then general materials and we add aluminum alloy from there. From above we can see the material properties of the aluminum alloy.

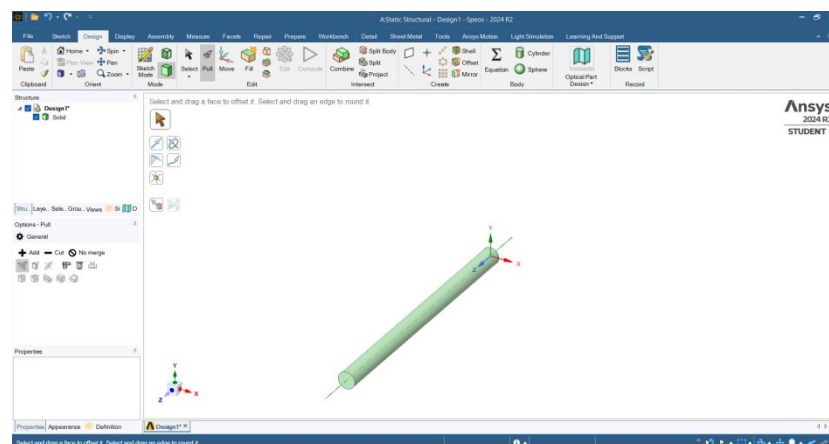


Figure 2 Geometry is drawn with respect to design dimensions

After adding and saving the engineering data we move with geometry section shown in Figure 2. When we click it opens a drawing program related with geometry. We draw a 2D circle with 10mm radius but we pull it 150mm to make it 3D. After that we also save the geometry and move on with “Model” section.

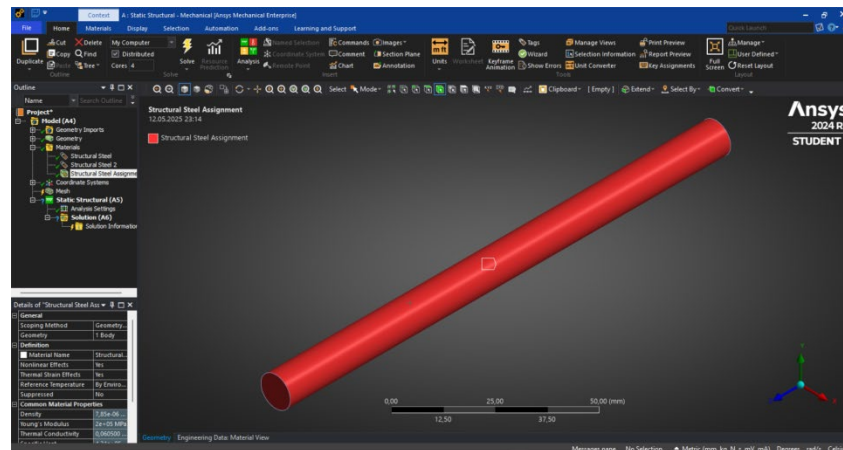


Figure 3 Structural Steel material is assigned to whole body

When we open the model section we see our cylindrical shape. First thing we need to do is go Materials section and select the aluminum alloy > right click on it > create material assignment and with these steps we'll assign aluminum alloy as our material. (Shown in Figure 3)

After that we right click on mesh and click generate mesh.

When done with meshing, we click static structural and click right on it and we select pin support from there and we apply it to one edge's surface. Then we right click static structural again and this time we are applying force but we select force type with magnitudes and we apply a -10N force on -Y axis.

After all steps done it'll look like the Figure 4 below.

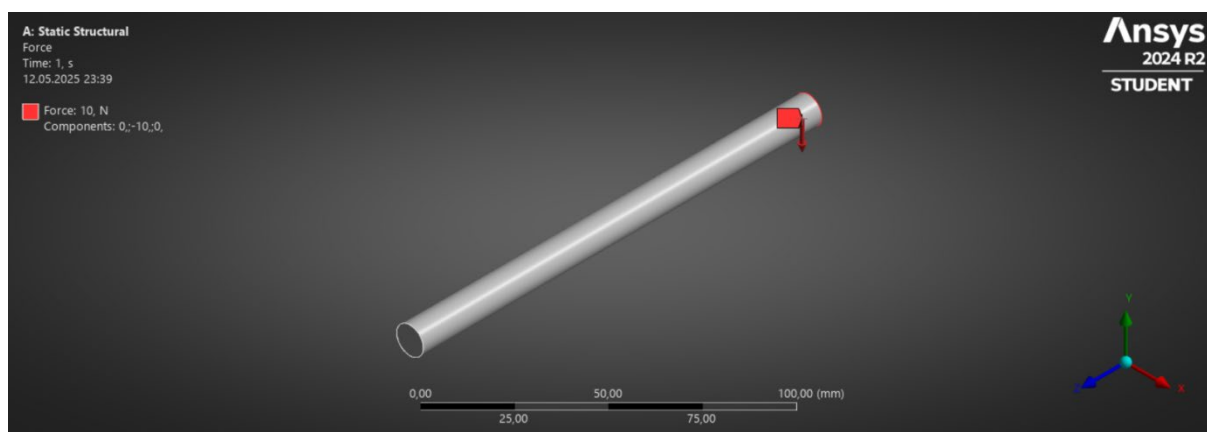


Figure 4 Mesh, fixed support, force and model done

Simulation

Now we will change material, size of the shaft and calculate the safety factor for different cases.

First, we look at how the result is different when we change material with same dimension.

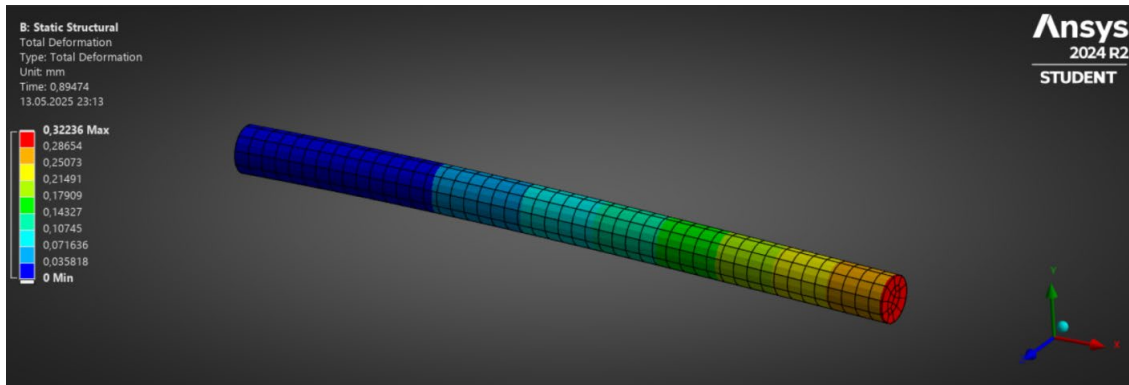


Figure 5 Deflection result for aluminum alloy

We have maximum deflection 0.32236mm at the end of the shaft which is 10N load is applied on -Y axis. (Shown in Figure 5)

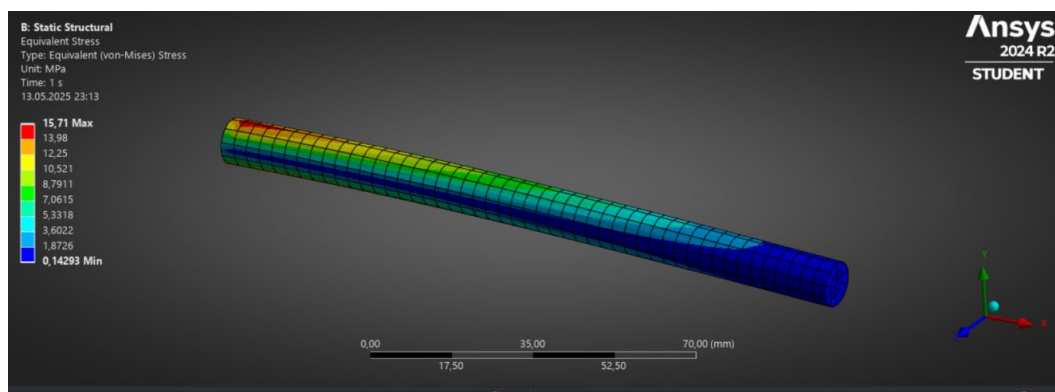


Figure 6 Von misses stress result for aluminum alloy

We have maximum stress (Von Mises) 15.71 MPa at the support and it reduces when we go to the load point. (shown in Figure 6)

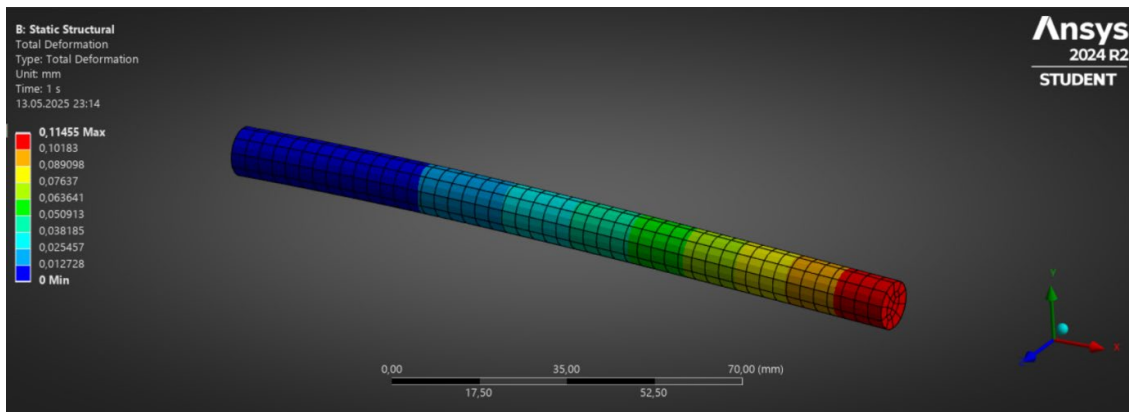


Figure 7 Deflection result for structural steel

In Figure 7 we change the material from aluminum alloy to structural steel same amount of load and same support. Only difference is the material. This time we have 0.11455 mm deflection which is less than aluminum alloy.

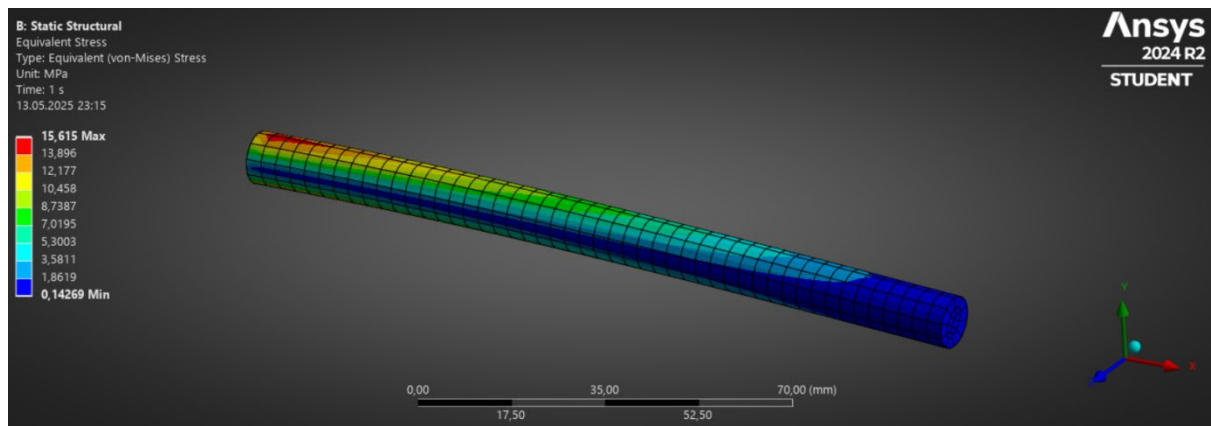


Figure 8 Von misses stress result for structural steel

For structural steel we have stress 15.615 MPa a little larger than the aluminum alloy. The stress points are very a look-like.

Table 2 Comparison between the properties of 2 different materials for same diameter

Property	Structural Steel	Aluminum Alloy (6061-T6)
Material	Structural Steel	6061-T6 Aluminum Alloy
Elastic Modulus (E)	200 GPa	69 GPa
Yield Strength	~250 MPa	276 MPa
Applied Force (F)	10 N	10 N
Shaft Length (L)	150 mm	150 mm
Diameter (d)	10 mm	10 mm
Max Deflection (ANSYS)	0.11455 mm	0.32236 mm
Von Mises Stress (ANSYS)	15.615 MPa	15.71 MPa
Safety Factor	15	15

Table 2 explains the result for same diameter and same length shaft but different material and different material properties.

Now we are going to change the size of the shaft.

10mm > 15mm diameter change

150mm > 120mm length change

We will obtain result again for the both structural steel and aluminum alloy and we'll also calculate new safety factor numbers.

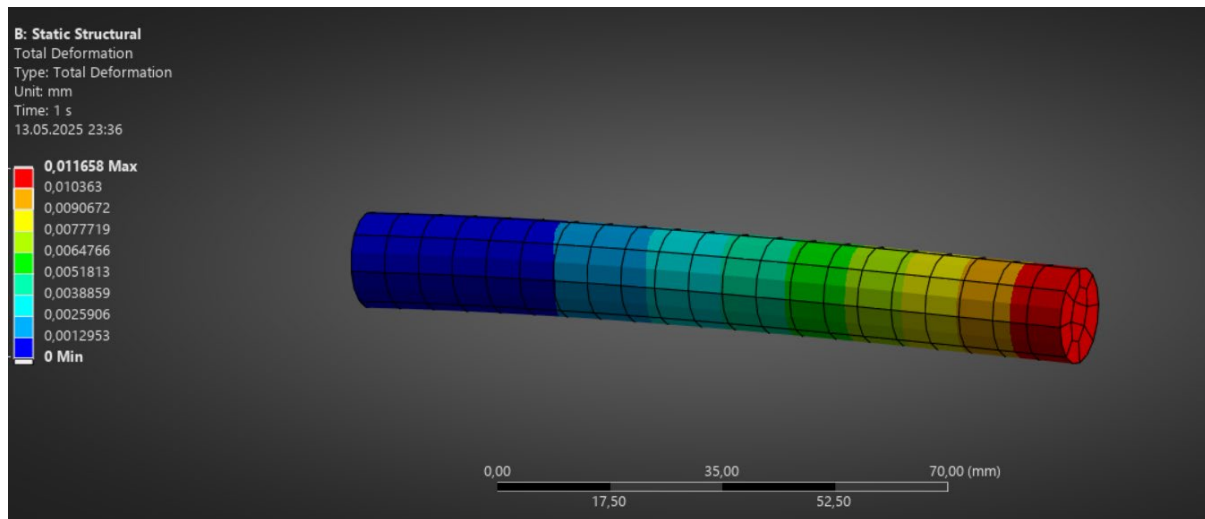


Figure 9 Displacement Result for Structural Steel (New dimensions)

In Figure 9 We did same steps as before. Like engineering data > adding aluminum alloy to the data library. To be used later. After that we saved it and open Geometry section but this time we draw the shape with a larger diameter which is 15mm and less length which is 120mm. We assigned the structural steel material first and after that we generated mesh and we applied same fixed support to one and we again applied a load 10N in -Y axis downward.

When we change the dimensions deflection for structural steel shaft is changed. It became 0.011658mm which is nearly 10 times smaller than the other shaft dimension result for same structural steel material.

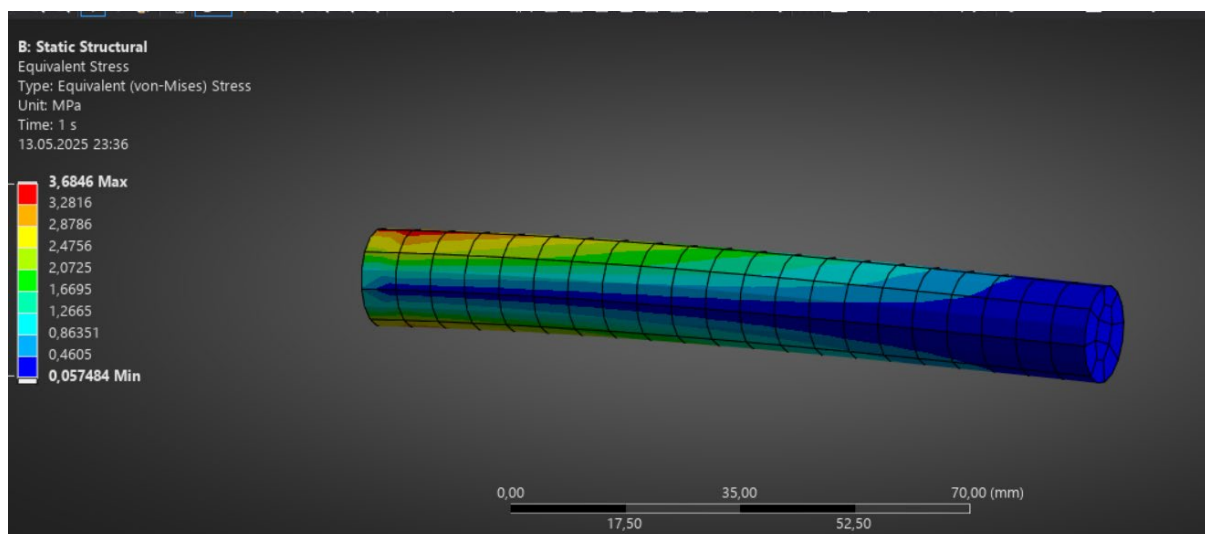


Figure 10 Von Misses Stress Results for Structural Steel (New dimensions)

Stress is equal to 3.6846 MPa (shown in Figure 10). Von mises stress equals mostly at the support point again. Stress is reduced nearly 4.5 times compared to other dimensions for same structural steel.

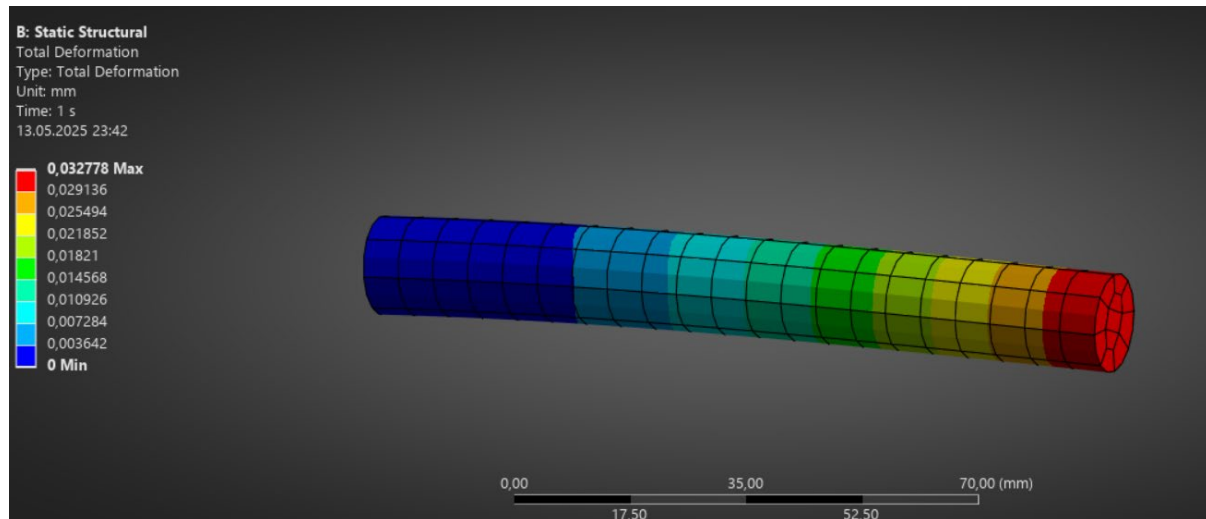


Figure 11 Displacement Result for Aluminum Alloy (New dimensions)

This time we do it with aluminum alloy again. Deflection result is equal to 0.032778mm again it is smaller nearly 10 times than the other aluminum alloy. (Shown in Figure 11)

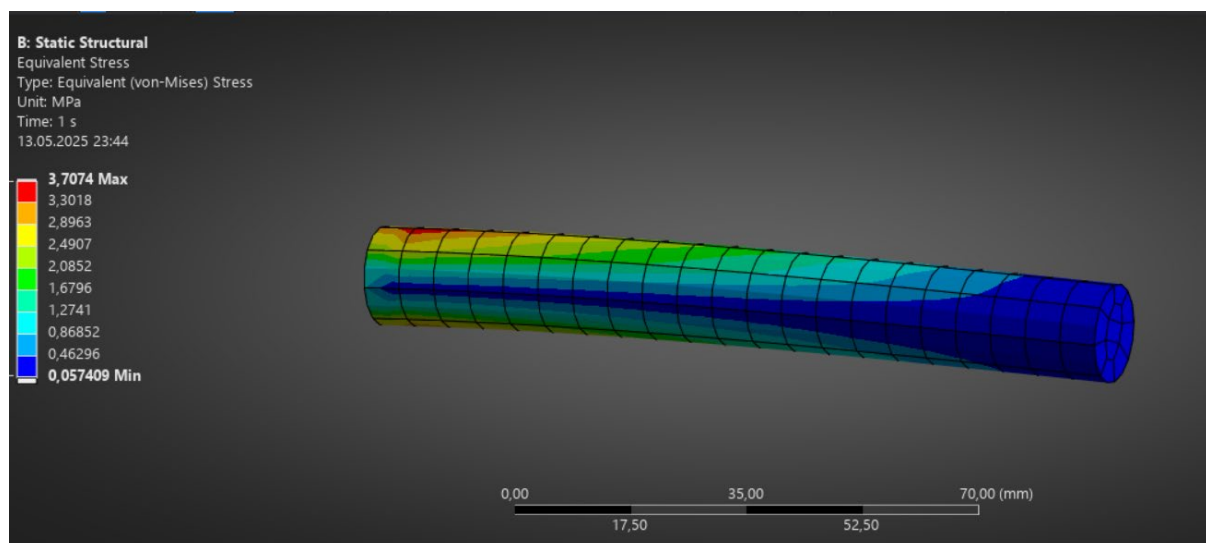


Figure 12 Von Misses Stress Results for Aluminum Alloy (New dimensions)

Stress is equal to 3.7074 MPa. Nearly 4.5 times less than the other dimensions for aluminum alloy material. Stress points are again mostly occurs at the support point. (Shown in Figure 12)

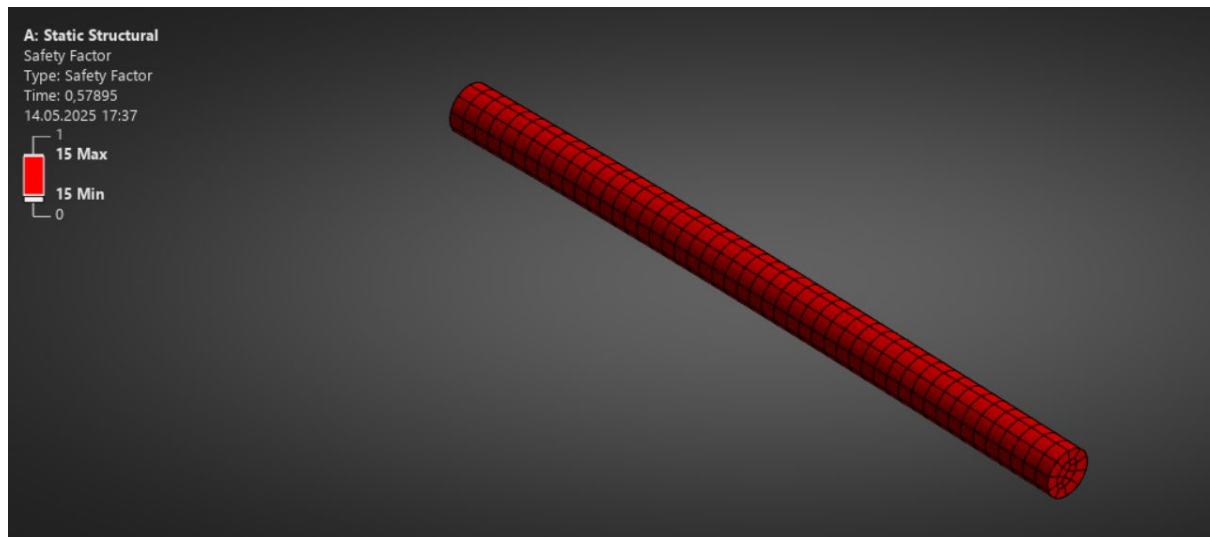


Figure 13 Safety Factor Result for 10mm diameter and 150mm length structural steel shaft

Result is same for 10mm diameter and 150mm length aluminum alloy shaft

We right click on the solution > stress tools > safety factor

And we solve for it and we can calculate and simulate the amount of safety factor for both materials and both shapes. It occurs to be same shown as Figure 13 and Figure 14

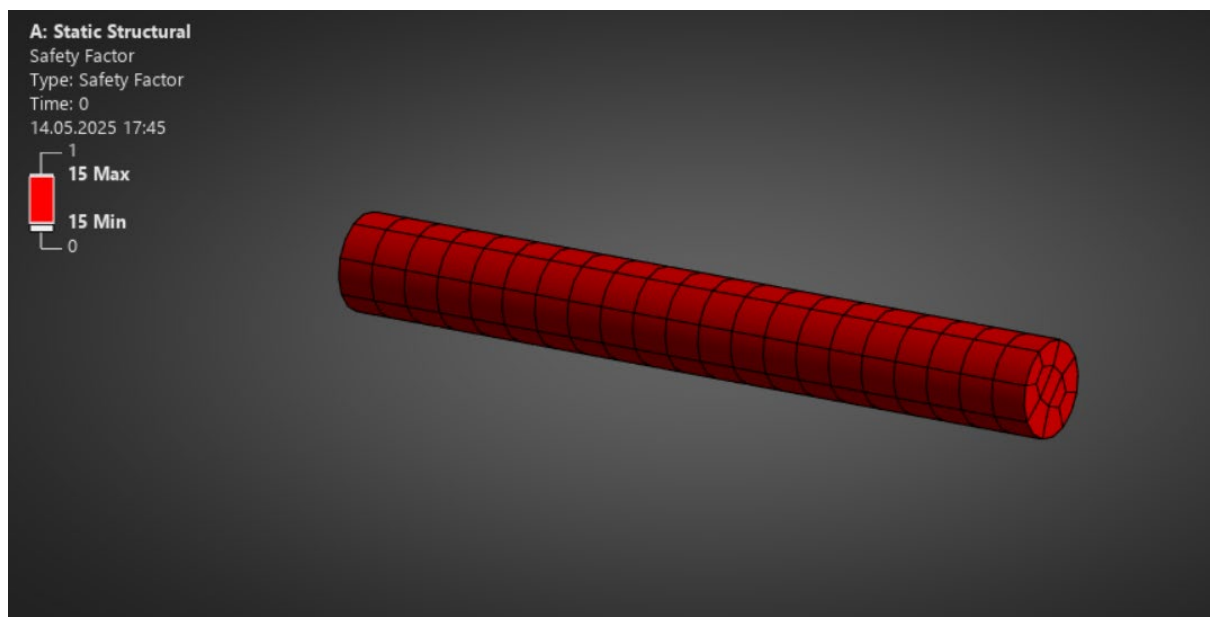


Figure 14 Safety Factor Result for 15mm diameter and 120mm length structural steel shaft

Result is same for 15mm diameter and 120mm length aluminum alloy shaft

Table 3 Displacement, stress and safety factor comparison between different diameter – length and materials

Material	Diameter (mm)	Length (mm)	Displacement (mm)	Stress (MPa)	Safety Factor
Steel	10	150	0.11455	15.615	15
Steel	15	120	0.01166	3.6846	15
Aluminum	10	150	0.32236	15.71	15
Aluminum	15	120	0.03278	3.7074	15

Table 4 Difference between displacement, stress and safety factor for different materials

Material	Displacement Difference (mm)	Stress Difference (MPa)	Safety Factor Difference
Steel	0.10289	11.93	0
Aluminum	0.28958	12.00	0

Table 3 and Table 4 shows the comparison between dimensions-material types, deflections and the number of stresses and safety factors. Also, we can see the difference between them.

Results

This section presents the outcomes of the static structural simulations performed on mechanical shafts using ANSYS Workbench. The analysis focused on assessing deformation, von Mises stress, and safety factors under different materials and geometrical configurations. Two primary comparative studies were conducted: one assessing the impact of material properties on shaft behavior and the other evaluating the effect of geometrical modifications.

Material Comparison: Constant Geometry

The first part of the analysis examined the shaft's performance when the material was changed while keeping the geometry constant. The shaft had a diameter of 10 mm and a length of 150 mm. The materials used were Structural Steel (AISI 1045) and Aluminum Alloy (6061-T6), both subjected to a static force of 10 N applied at the free end of a cantilevered shaft.

Simulation results for Structural Steel:

- Maximum deflection: 0.11455 mm
- Von Mises stress: 15.615 MPa
- Safety factor: 15

Simulation results for Aluminum Alloy (6061-T6):

- Maximum deflection: 0.32236 mm
- Von Mises stress: 15.71 MPa
- Safety factor: 15

These results indicate that while the stress values were nearly identical, the aluminum shaft exhibited significantly higher deflection due to its lower Young's Modulus (69 GPa) compared to structural steel (200 GPa). However, both materials maintained a high safety factor, suggesting that they are structurally capable under the given loading conditions.

Geometrical Modification: Material Constant and Variable

In the second part of the analysis, the shaft geometry was modified to a diameter of 15 mm and a length of 120 mm. The same materials (Structural Steel and Aluminum Alloy) were evaluated under the same loading conditions (10 N applied force at the free end).

Simulation results for Structural Steel (Modified Geometry):

- Maximum deflection: 0.01166 mm
- Von Mises stress: 3.6846 MPa
- Safety factor: 15

Simulation results for Aluminum Alloy (Modified Geometry):

- Maximum deflection: 0.03278 mm
- Von Mises stress: 3.7074 MPa
- Safety factor: 15

The modified geometry significantly reduced both displacement and stress levels for both materials. The shorter length and increased diameter increased the shaft's stiffness and reduced bending moments, which directly contributed to the reduction in deformation. The safety factor remained consistently high across both materials, reaffirming structural adequacy.

Discussion

The result obtained in this study provides comprehensive insights into how material property and geometric have effect on the performance of shafts. While the safety factor is maintained at for both materials, the stiffness between steel and aluminum resulted in different deformation behavior under the same loading condition. The aluminum shaft's deflection was greater than steel shaft's deflection.

From the results obtained, the geometric modifications have showed its crucial impact, with the increased diameter from 10mm to 15mm and reducing length of 150mm to 120mm, we observed a significant improvement in stiffness. The significant increased stiffness showed how dimension adjustment can effectively compensate for material's limitations.

The constant reduction of stress in both materials at 76% indicates that geometric optimization is advantageous for all load-bearing parts. These finding shows that it is crucial in practice for design engineers. The stress concatenation patterns reveal that at the support location, established mechanical principles are validated while FEA's precision in prediction is seen in the critical failure zones. The information obtained has greatly simplified the decision-making process for material selection regarding the trade-off between aluminum's weight advantage and steel's rigidity.

Conclusion

This report has successfully illustrated the static analysis of the mechanical shaft counted through FEA (finite element analysis) by using ANSYS Workbench, this study is focused on evaluating stress distribution, deformation and safety factor in different geometry configuration for 2 chosen materials, structural steel (AISI1045) and aluminum alloy(6061-T6). The results carried out by using ANSYS shows that steel has a better stiffness(200GPa) compared to aluminum (GPa) which reduces the deflection by 65%.Both material maintained safe stress levels of (15.6MPa) and safety factor of 15. When diameter is increased from 10mm to 15 mm and reducing the length from 150mm to 120mm will decrease the deflection by 90% and the stress by 76% for both material , this shows that changing in geometric will have a significant effect on the behavior of material. The FEA results we obtained have effectively identified the stress concentrations at support points, validating the method's reliability for design analysis. This report has successfully demonstrated FEA's value in optimizing shaft design while minimizing cost of making a physical prototype. This project has successfully enhanced students' ability and understanding of how to apply FEA in practical engineer design.

References

- [1] Abidov K, Dadadjanov K, Sobitov S. Static modes of an electromagnetic working shaft (EMWSh) and an electro-electromagnetic working shaft (EEMWSh) with the inclusion of a capacitor in the rotor circuit of the motors. E3S Web Conf 2023;410:02058. <https://doi.org/10.1051/e3sconf/202341002058>.
- [2] Alymenko DN, Soloviev VA, Aptukov VN, Kotlyar EK. Lightweight support types for reinforcement of mining openings adjacent to shafts in salt rocks. Gorn Zhurnal 2019;42–5. <https://doi.org/10.17580/gzh.2019.05.07>.
- [3] Ašonja Aleksandar, Zivoslav Adamovic NJ. Analysis of reliability of cardan shafts based on condition diagnostics of bearing assembly in cardan joints 2013.
- [4] Ašonja A, , Eleonora Desnica IP. ANALYSIS OF THE STATIC BEHAVIOR OF THE SHAFT BASED ON FINITE ELEMENT METHOD UNDER EFFECT OF DIFFERENT VARIANTS OF LOAD. 2016.
- [5] Palinkaš I. Application of computer technologies (CAD CAM systems) for quality improvement of education 2016.
- [6] Asonja A, Desnica E. Reliability of agriculture universal joint shafts based on temperature measuring in universal joint bearing assemblies. Spanish J Agric Res 2014;13:e0201. <https://doi.org/10.5424/sjar/2015131-6371>.
- [7] Buckley RM, Jardine RJ, Kontoe S, Barbosa P, Schroeder FC. Full-scale observations of dynamic and static axial responses of offshore piles driven in chalk and tills. Géotechnique 2020;70:657–81. <https://doi.org/10.1680/jgeot.19.TI.001>.
- [8] Choi J-W, Han S-H, Lee K-H. Structural analysis and optimization of an automotive propeller shaft. Adv Mech Eng 2021;13. <https://doi.org/10.1177/16878140211053173>.
- [9] Ivanovic L. Cross Shaft Design From the Aspect of Capacity 2011.

- [10] Diulin DA, Prushak VY, Gegedesh MG. Analysis of the stress-strain state of problematic sections of the shaft of the mine using computer simulation. *Dokl Natl Acad Sci Belarus* 2023;67:322–30. <https://doi.org/10.29235/1561-8323-2023-67-4-322-330>.
- [11] Đorđević Z. Analysis of the Influence of the Fibre Type on Static and Dynamic Characteristics of Composite Shafts 2011.
- [12] Maksimovic S. Dynamic analysis of hybrid aluminum-composite shafts 2008.
- [13] Hosseini-Tabatabaei M-R, Alfouneh M. Closed-form equations for statically analyzing shaft-gear systems. *Meccanica* 2024;59:921–38. <https://doi.org/10.1007/s11012-024-01803-4>.
- [14] Jweeg MJ, Alnomani SN, Mohammad SK. Dynamic analysis of a rotating stepped shaft with and without defects. *IOP Conf Ser Mater Sci Eng* 2020;671:012004. <https://doi.org/10.1088/1757-899X/671/1/012004>.
- [15] Karasev MA, Buslova MA, Vilner MA, Nguyen TT. Method for predicting the stress-strain state of the vertical shaft lining at the drift landing section in saliferous rocks. *J Min Inst* 2019;240:628–37. <https://doi.org/10.31897/pmi.2019.6.628>.
- [16] Liang P, Li X, Guo F, Zhang X, Jiang F. Influence of Shaft Scratch on Static and Transient Behavior of Water-Lubricated Bearing. *J Tribol* 2024;146. <https://doi.org/10.1115/1.4062814>.
- [17] Ma R, Jin H, Yi Y, Yang J, Fan X. Design and combustion characteristics analysis of a static shaft turbofan engine. *Int J Turbo Jet-Engines* 2024;41:539–50. <https://doi.org/10.1515/tjj-2023-0057>.
- [18] Nanda J, Parhi DR. Theoretical Analysis of the Shaft. *Adv Fuzzy Syst* 2013;2013:1–11. <https://doi.org/10.1155/2013/392470>.
- [19] Oñate E. Structural Analysis with the Finite Element Method. Dordrecht: Springer

- Netherlands; 2009. <https://doi.org/10.1007/978-1-4020-8733-2>.
- [20] Desnica E. Reliability model of bearing assembly on an agricultural cardan shaft 2015.
 - [21] Rasool S. Three Dimensional Modelling and Stress Analysis of a Powered Single Acting Disc Harrow Using FEA. 2017.
 - [22] Schillinger D, Ruess M. The Finite Cell Method: A Review in the Context of Higher-Order Structural Analysis of CAD and Image-Based Geometric Models. *Arch Comput Methods Eng* 2015;22:391–455. <https://doi.org/10.1007/s11831-014-9115-y>.
 - [23] Siostrzonek T. The Mine Shaft Energy Storage System—Implementation Threats and Opportunities. *Energies* 2023;16:5615. <https://doi.org/10.3390/en16155615>.
 - [24] Aleksandar Ašonja ZM. Static behaviour analysis of centrifugal pump shaft 2004.
 - [25] Letić, D., Branko Davidovic BR. The high - Performance algorithm of the computer methods at the establishing of the states of stress of the brake mechanism by the finite element method (FEM) 2012.
 - [26] Zhang J, Ma Z, Wang X, Zhang L. The stability analysis and nonlinear vibration control of shaft system for hydraulic generating set under multi-source excitation considering effects of dynamic and static eccentricities. *Commun Nonlinear Sci Numer Simul* 2023;121:107230. <https://doi.org/10.1016/j.cnsns.2023.107230>.
 - [27] Zotov A, Sviridov A, Valeev A. Stabilization of the Torque on the Shaft of a Statically Balanced Crank Mechanism. 2020 Int. Conf. Dyn. Vibroacoustics Mach., IEEE; 2020, p. 1–7. <https://doi.org/10.1109/DVM49764.2020.9243882>.
 - [28] Spotts MF, Shoup TE. *Design of Machine Elements*. Prentice Hall; 2003.
 - [29] Budynas RG, Nisbett JK. *Shigley's Mechanical Engineering Design*. McGraw-Hill Education; 2020.
 - [30] Moaveni S. *Finite Element Analysis Using ANSYS*. Cengage Learning; 2015.
 - [31] Lawrence KL. *ANSYS Workbench Tutorial*. SDC Publications; 2012.

- [32] Deshmukh SS, Kulkarni PS. Finite Element Analysis of Shaft of Centrifugal Pump. IOSR J Mech Civ Eng 2013;7:37–42.
- [33] Kadam MS, Kulkarni SS. Stress Analysis of Spline Shaft Using Finite Element Method and Its Experimental Validation. Int J Mech Eng Robot Res 2015;4:160–5.
- [34] Khan A, Singh V. Design and Static Structural Analysis of Composite Drive Shaft Using Finite Element Analysis Method. Int J Res Anal Rev 2019;6:86–92.
- [35] DIN 743 – Calculation of load capacity of shafts and axles 2012.
- [36] ASME B106.1M – Design of Transmission Shafting 1985.